Probability Measures

Math Foundations – Node 19

First, an example

Before the axioms, the running picture: a *fair six-sided die*. Take $\Omega = \{1,2,3,4,5,6\}$ and $\mathcal{F} = 2^{\Omega}$ (every subset is an event). Define $P(\{i\}) = 1/6$ for each face i, and extend by addition: $P(A) = |A|/6$. Then the three Kolmogorov axioms are visible at a glance: (K1) $P(A) \geq 0$ since $|A| \geq 0$; (K2) $P(\Omega) = 6/6 = 1$; (K3, finite case) for disjoint events the counts add, so e.g. $P(\{1,2\}) = 2/6 = 1/6 + 1/6 = P(\{1\}) + P(\{2\})$. Every formal manoeuvre below is just a careful version of what already works for the die.

1 Setting

Throughout, Ω is a non-empty set (the *sample space*) and $\mathcal{F} \subseteq 2^{\Omega}$ is a σ-algebra on Ω in the sense of node 18: $\Omega \in \mathcal{F}$, $\mathcal{F}$ is closed under complements, and $\mathcal{F}$ is closed under countable unions. Elements of $\mathcal{F}$ are called *events*.

2 Definitions

Definition 1 (Probability measure)**.** A *probability measure* on $(\Omega, \mathcal{F})$ is a set function $P : \mathcal{F} \to [0,1]$ satisfying the three *Kolmogorov axioms*:

(K1) *Non-negativity:* $P(A) \geq 0$ for every $A \in \mathcal{F}$.

(K2) *Normalisation:* $P(\Omega) = 1$.

(K3) *Countable additivity (σ-additivity):* for every sequence $\{A_n\}_{n\geq 1} \subseteq \mathcal{F}$ of pairwise disjoint events,

$$P\left(\bigcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} P(A_n).$$

Read aloud: "P of the union from $n = 1$ to infinity of A_n equals the sum from $n = 1$ to infinity of P of A_n."

Definition 2 (Probability space)**.** A *probability space* is a triple $(\Omega, \mathcal{F}, P)$ where Ω is a sample space, $\mathcal{F}$ is a σ-algebra on Ω, and P is a probability measure on $(\Omega, \mathcal{F})$.

Remark 3. From (K1)–(K3) one immediately obtains $P(\emptyset) = 0$ (take all $A_n = \emptyset$ in (K3); the sum $\sum P(\emptyset)$ converges only if each term vanishes) and finite additivity $P(A \sqcup B) = P(A) + P(B)$ by padding with empty sets.

3 The inclusion–exclusion identity

Theorem 4 (Inclusion–exclusion, two-set form)**.** *Let* $(\Omega, \mathcal{F}, P)$ *be a probability space. For every* $A, B \in \mathcal{F}$*,*

$$P(A \cup B) \;=\; P(A) + P(B) - P(A \cap B).$$

Read aloud: *"P of A union B equals P of A plus P of B minus P of A intersect B."*

Proof. We decompose $A \cup B$ and B into disjoint pieces and apply finite additivity, which follows from (K3) as noted above.

Step 1: $A \cup B = A \sqcup (B \setminus A)$ **disjointly.** Every $\omega \in A \cup B$ is in A or in $B \setminus A := B \cap A^c$, and these two sets are disjoint by construction. Both lie in $\mathcal{F}$: $A \in \mathcal{F}$ by hypothesis, and $B \setminus A = B \cap A^c \in \mathcal{F}$ since $\mathcal{F}$ is closed under complements and finite intersections. By finite additivity,

$$P(A \cup B) \;=\; P(A) + P(B \setminus A). \tag{1}$$

Read aloud: "P of A union B equals P of A plus P of B minus A."

Step 2: $B = (B \cap A) \sqcup (B \setminus A)$ **disjointly.** Every $\omega \in B$ either lies in A (hence in $B \cap A$) or does not (hence in $B \setminus A$), and these two sets are disjoint. Both lie in $\mathcal{F}$ for the same reasons as in Step 1. By finite additivity,

$$P(B) \;=\; P(B \cap A) + P(B \setminus A),$$

Read aloud: "P of B equals P of B intersect A plus P of B minus A." and rearranging gives

$$P(B \setminus A) \;=\; P(B) - P(A \cap B). \tag{2}$$

Read aloud: "P of B minus A equals P of B minus P of A intersect B." (Note $P(B \cap A) = P(A \cap B)$ since intersection is commutative.)

Step 3: combine. Substituting (2) into (1),

$$P(A \cup B) \;=\; P(A) + \big(P(B) - P(A \cap B)\big) \;=\; P(A) + P(B) - P(A \cap B). \qquad \square$$

Read aloud: "P of A union B equals P of A plus the quantity P of B minus P of A intersect B, which equals P of A plus P of B minus P of A intersect B."

Remark 5. The general n-set inclusion–exclusion formula

$$P\left(\bigcup_{i=1}^{n} A_i\right) \;=\; \sum_{k=1}^{n} (-1)^{k+1} \sum_{1 \le i_1 < \cdots < i_k \le n} P(A_{i_1} \cap \cdots \cap A_{i_k})$$

Read aloud: "P of the union from $i = 1$ to n of A_i equals the sum from $k = 1$ to n of negative one to the k plus one, times the sum over all k-element index subsets $i_1 < \cdots < i_k$ of P of A_{i_1} intersect dot dot dot intersect A_{i_k}." follows by induction on n, with Theorem 4 as the base case.

4 An immediate corollary

Proposition 6 (Boole's inequality)**.** *For every countable collection* $\{A_n\}_{n \ge 1} \subseteq \mathcal{F}$*,*

$$P\left(\bigcup_{n=1}^{\infty} A_n\right) \;\le\; \sum_{n=1}^{\infty} P(A_n).$$

Read aloud: *"P of the union from* $n = 1$ *to infinity of* A_n *is less than or equal to the sum from* $n = 1$ *to infinity of P of* A_n*."*

Proof. Set $B_1 = A_1$ and $B_n = A_n \setminus \bigcup_{k<n} A_k$ for $n \geq 2$. The B_n lie in $\mathcal{F}$, are pairwise disjoint, and satisfy $\bigcup_n B_n = \bigcup_n A_n$ with $B_n \subseteq A_n$. By (K3) and monotonicity (itself a one-line consequence of Theorem 4 applied with $B \supseteq A$),

$$P\left(\bigcup_n A_n\right) = \sum_n P(B_n) \leq \sum_n P(A_n). \qquad \square$$

Read aloud: "P of the union over n of A_n equals the sum over n of P of B_n, which is less than or equal to the sum over n of P of A_n."